ONE DIMENTIONAL ARRAY

1. Second largest element

```
#include <stdio.h>
int main()
{
    int n;
    scanf("%d", &n);
    int arr[n];
  for(int i = 0; i < n; i++)
  {
    scanf("%d", &arr[i]);
  }
  int largest = arr[0];
  int second = arr[0];
  for(int i = 0; i < n; i++)
 {
    if(arr[i] > largest)
    {
        second = largest;
        largest = arr[i];
    }
    else if(arr[i] > second && arr[i] != largest)
    {
      second = arr[i];
```

```
        }
      }
    printf("Second Largest = %d", second);
   return 0;
   }
```

2.Missing Number In Array

```
  #include <stdio.h>
 int main()
{
 int n;
 scanf("%d", &n);
 int arr[n];
 for(int i=0;i<n; i++)
 {
    scanf("%d",&arr[i]);
 }
 int xor1=0;
for(int i=1;i<=n+1;i++)
{
   xor1^=1;
}
for(int i=0;i<n; i++)
{
   xor2^=arr[i];
}
```

```
 printf("%d", xor1^xor2);
    return 0;
}
```

3.Roate An Array

```
#include <stdio.h>
 int main()
{
  int n;
  scanf("%d",&n);
  int k;
  scanf("%d",&k);
  int arr[n];
  for(int i=0;i<n;i++)
  {
     scanf("%d",&arr[i]);
  }
  int last;
  for(int i=0;i<k;i++)
  {
     last=arr[0];
     for(int i=0;i<n-1;i++)
     {
        arr[i]=arr[i+1];
     }
     arr[n-1]=last;
```

```
   }
 for(int j=0;j<n;j++)
   {
      printf("% d",arr[j]);
   }
   return 0;
}
```
4.Duplicate In an Array
```
     #include <stdio.h>
     int main()
{
   int n;
   scanf("%d",&n);
   int arr[n];
   for(int i=0;i<n;i++)
   {
      scanf("%d",&arr[i]);
   }

   for(int i=0;i<n;i++)
   {
      for(int j=i+1;j<n;j++)
      {
        if(arr[i]==arr[j])
        {
```

```
            printf(" %d",arr[i]);
            break;
          }
        }
    }
    return 0;
}
```
5.Find leaders In An Array
```
    #include <stdio.h>
   int main()
{
  int n;
  scanf("%d",&n);
  int arr[n];
  for(int i=0;i<n;i++)
  {
     scanf("%d",&arr[i]);
  }
  int max=arr[n-1];
  printf(" %d",max);
  for(int i=n-2;i>=0;i--)
  {
     if(arr[i]>max)
     {
        max=arr[i];
```

```
        printf(" %d",max);
    }
}
return 0;
}
```

6.Find The Union Of Two Arrays

```
    #include <stdio.h>
  int main()
{
  int n1,n2;
  scanf("%d %d",&n1,&n2);
  int arr1[n1];
  int arr2[n2];
  for(int i=0;i<n1;i++)
  {
     scanf("%d",&arr1[i]);
  }
  for(int j=0;j<n2;j++)
  {
     scanf("%d",&arr2[j]);
  }
  int unionarr[20];
  int k=0;
  for(int i=0;i<n1;i++)
  {
```

```
      unionarr[k++]=arr1[i];
    }
    for(int j=0;j<n2;j++)
    {
       int found=0;
       for(int i=0;i<k;i++)
       {
          if(unionarr[j]==arr2[i])
          {
           found=1;
           break;
          }
       }
       if(found==0)
       {
       unionarr[k++]=arr2[j];
       }
    }
 for(int h=0;h<k;h++)
 {
    printf(" %d",unionarr[h]);
 }
   return 0;
}
```

7.Find The Intersection Of Two Arrays

```
#include <stdio.h>
int main()
{
  int n1,n2;
  scanf("%d %d",&n1,&n2);
  int arr1[n1];
  int arr2[n2];
  for(int i=0;i<n1;i++)
  {
     scanf("%d",&arr1[i]);
  }
  for(int i=0;i<n2;i++)
  {
     scanf("%d",&arr2[i]);
  }
  int intersecarr[20];
  int k=0;
  for(int i=0;i<n1;i++)
  {
     for(int j=0;j<n2;j++)
     {
        if(arr1[i]==arr2[j])
        {
           intersecarr[k]=arr1[i];
           k++;
```

```
        }
      }
    }
    for(int i=0;i<k;i++)
    {
      printf(" %d",intersecarr[i]);
    }
    return 0;
}
```

8.Find The First Repeating Elements

```
    #include <stdio.h>

int main()
{
    int n;
    scanf("%d",&n);
    int arr[n];
    for(int i=0;i<n;i++)
    {
        scanf("%d",&arr[i]);
    }
    for(int i=0;i<n;i++)
    {
        for(int j=i+1;j<n;j++)
        {
```

```
            if(arr[i]==arr[j])
            {
                printf("%d",arr[i]);
                return 0;
            }
        }
    }
}
```

9.Find The First Non-Repeating Element

```
  #include<stdio.h>
int main()
{
  int n;
  scanf("%d",&n);
  int arr[n];
  for(int i=0;i<n;i++)
  {
      scanf("%d",&arr[i]);
  }
  for(int i=0;i<n;i++)
  {
      int count=0;
      for(int j=0;j<n;j++)
      {
          if(arr[i]==arr[j])
```

```
        {
          count++;
        }
      }
      if(count==1)
      {
        printf("%d",arr[i]);
        break;
      }

    }
 return 0;
}
```

10.Maximum Consecutives Ones

```
      #include <stdio.h>
int main()
{
   int n;
   scanf("%d",&n);
   int arr[n];
   for(int i=0;i<n;i++)
   {
      scanf("%d",&arr[i]);
   }
   int count=0;
```

```
    int maxcount=0;
    for(int i=0;i<n;i++)
    {
       if(arr[i]==1)
       {
          count++;
           if(count>maxcount)
            {
              maxcount=count;
            }
       }
       else
      {
         count=0;
       }

    }
    printf("%d",maxcount);
    return 0;
}
```

11.Largest Sum Contiguous Subarray Of Size K

```
      #include <stdio.h>
#include <stdlib.h>
int main()
{
```

```
int n,k;
scanf("%d %d",&n,&k);
int *arr;
arr=(int*)calloc(n,sizeof(int));
int sum=0;
int maxsum=0;
for(int i=0;i<n;i++)
{
   scanf("%d",&arr[i]);
}
for(int i=0;i<=n-k;i++)
{
   for(int j=i;j<i+k;j++)
   {
      sum+=arr[j];
   }
   if(sum>maxsum)
   {
      maxsum=sum;
   }
   else
   {
      sum=0;
   }
}
```

```
    printf(" %d",maxsum);
    return 0;
}
```

12.Count Inversion In An Array

```
        #include <stdio.h>
#include <stdlib.h>
void countinversion(int *arr,int n)
{
    int count=0;
    for(int i=0;i<n;i++)
    {
        for(int j=0;j<n;j++)
        {
            if(i<j&&arr[i]>arr[j])
            {
                count++;
            }
        }
    }
    printf("%d",count);
}
int main()
{
    int n;
    scanf("%d",&n);
```

```
    int *arr;
    arr=(int*)calloc(n,sizeof(int));
    for(int i=0;i<n;i++)
    {
       scanf("%d",&arr[i]);
    }
     countinversion(arr,n);
     free(arr);
    return 0;
}
```

13.Find Pair With Given Sum

```
      #include <stdio.h>
#include <stdlib.h>
void pairsum(int *arr, int n, int sum)
{
    for(int i = 0; i < n; i++)
    {
        for(int j = i + 1; j < n; j++)
        {
            if(arr[i] + arr[j] == sum)
            {
                printf("YES");
                return;
            }
        }
```

```
    }
    printf("NO");
}
int main()
{
    int n, sum;
    scanf("%d %d", &n, &sum);
    int *arr = (int *)calloc(n, sizeof(int));
    for(int i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }
    pairsum(arr, n, sum);
    return 0;
}
```

14.Maximum Difference

```
      #include <stdio.h>
#include <stdlib.h>
void difference(int *arr,int n)
{
    int min=arr[0];
    int maxdiff=arr[1]-arr[0];
    for(int i=1;i<n;i++)
    {
        if(arr[i]-min>maxdiff)
```

```
        {
           maxdiff=arr[i]-min;
        }
        if(arr[i]<min)
        {
           min=arr[i];
        }
    }
    printf("%d",maxdiff);
}
int main()
{
   int n;
   scanf("%d",&n);
   int *arr;
   arr=(int *)calloc(n,sizeof(int));
   for(int i=0;i<n;i++)
   {
     scanf("%d",&arr[i]);
   }
   difference(arr,n);
    return 0;
}
```

15.Product Of Array Except Self

```
        #include <stdio.h>
```

```
#include <stdlib.h>
void product(int *arr,int n)
{
   for(int i=0;i<n;i++)
   {
      int prod=1;
      for(int j=0;j<n;j++)
      {
         if(arr[i]==arr[j])
         {
            continue;
         }
         prod*=arr[j];
      }
      printf(" %d",prod);
   }
}
int main()
{
   int n;
   scanf("%d",&n);
   int *arr;
   arr=(int *)calloc(n,sizeof(int));
   for(int i=0;i<n;i++)
   {
```

```
    scanf("%d",&arr[i]);
  }
  product(arr,n);
  return 0;
}
```

16.Chech If Array Is Sorted

```
    #include <stdio.h>
int main()
{
  int n;
  scanf("%d",&n);
 int arr[n];
 for(int i=0;i<n;i++)
 {
   scanf("%d",&arr[i]);
 }
 int i=0;
 int flag=1;
 while(i<n-1)
 {
   if(arr[i]>=arr[i+1])
     {
       flag=0;
       break;
     }
```

```
        i++;
    }
    if(flag==1)
    {
       printf("yes");
    }
    else
    {
       printf("no");
    }
     return 0;
}
```

17.Find The Smallest Positive Missing Number

```
#include <stdio.h>
int main()
{
    int n;
    scanf("%d", &n);
    int arr[n];
    for(int i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }
    for(int i = 0; i < n - 1; i++)
    {
```

```
        for(int j = i + 1; j < n; j++)
        {
            if(arr[i] > arr[j])
            {
                int temp = arr[i];
                arr[i] = arr[j];
                arr[j] = temp;
            }
        }
    }
    int missing = 1;
    for(int i = 0; i < n; i++)
    {
        if(arr[i] == missing)
        {
            missing++;
        }
    }
    printf("%d", missing);
    return 0;
}
```

18.Move All Zeros To The End

```
#include <stdio.h>
int main()
{
```

```
int n;
scanf("%d",&n);
int arr[n], brr[n];
for(int i=0;i<n;i++)
{
    scanf("%d",&arr[i]);
}
int k=0;
for(int i=0;i<n;i++)
{
    if(arr[i]!=0)
    {
        brr[k]=arr[i];
        k++;
    }
}
while(k<n)
{
    brr[k]=0;
    k++;
}
for(int i=0;i<n;i++)
{
    printf("%d ",brr[i]);
}
```

```
    return 0;
}
```

19.Find Frequency Of Each Element

```
  #include <stdio.h>
int main()
{
    int n;
    scanf("%d",&n);
    int arr[n];
    for(int i=0;i<n;i++)
    {
        scanf("%d",&arr[i]);
    }
    for(int i=0;i<n;i++)
    {
        int count=0;
        for(int j=0;j<n;j++)
        {
            if(arr[i]==arr[j])
            {
                count++;
            }
        }
        printf("%d = %d\n",arr[i],count);
    }
```

```
    return 0;
}
```

20.Longest Consecution Sequence

```
    #include <stdio.h>
int main()
{
    int n;
    scanf("%d",&n);
    int arr[n];
    for(int i=0;i<n;i++)
    {
        scanf("%d",&arr[i]);
    }
    for(int i=0;i<n-1;i++)
    {
        for(int j=i+1;j<n;j++)
        {
            if(arr[i] > arr[j])
            {
                int temp = arr[i];
                arr[i] = arr[j];
                arr[j] = temp;
            }
        }
    }
```

```
    int count = 1;
    int max = 1;
    for(int i=1;i<n;i++)
    {
        if(arr[i] == arr[i-1] + 1)
        {
            count++;
        }
        else
        {
            if(count > max)
                max = count;
            count = 1;
        }
    }
    if(count > max)
        max = count;
    printf("%d", max);
    return 0;
}
```